

Juzheng Zhang

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Education

University of Maryland, College Park

PH.D. STUDENT IN COMPUTER SCIENCE

College Park, MD

Aug 2024 – Present

- Advised by [Prof. Tom Goldstein](#); GPA: 4.0/4.0.
- Research on LLM and agent post-training, with interests in RL, continual learning, and self-improvement.

Shanghai Jiao Tong University

BACHELOR OF ENGINEERING IN INFORMATION ENGINEERING

Shanghai, China

Sep 2019 – Jun 2023

- Graduated with Honors (Top 1%); Overall GPA: 4.0/4.3; Rank: 1/152; Member of the Zhiyuan Honors Program.

Publications

Don't Mask the Environment: Observation Supervision Changes How Agents Explore Under RL

arXiv 2026

Juzheng Zhang, Disha Makhija, Manoj Ghuhan Arivazhagan, Vinayshekhar Bannihatti Kumar, Rashmi Gangadharaiah

Learning from Mixed Rollouts: Logit Fusion as a Bridge Between Imitation and Exploration

Blog Post

Juzheng Zhang, Abhimanyu Hans, John Kirchenbauer, Micah Goldblum, Ashwinee Panda, Tom Goldstein

LoRI: Reducing Cross-Task Interference in Multi-Task Low-Rank Adaptation

COLM 2025

Juzheng Zhang, Jiacheng You, Ashwinee Panda, Tom Goldstein

HIGHT: Hierarchical Graph Tokenization for Molecule-Language Alignment

ICML 2025

Yongqiang Chen, Quanming Yao, Juzheng Zhang, James Cheng, Yatao Bian

Heuristic Learning with Graph Neural Networks: A Unified Framework for Link Prediction

KDD 2024

Juzheng Zhang, Lanning Wei, Zhen Xu, Quanming Yao

Customized Subgraph Selection and Encoding for Drug-drug Interaction Prediction

NeurIPS 2024

Haotong Du, Quanming Yao, Juzheng Zhang, Yang Liu, Zhen Wang

UniMoT: Unified Molecule-Text Language Model with Discrete Token Representation

arXiv 2024

Juzheng Zhang, Yatao Bian, Yongqiang Chen, Quanming Yao

Experience

Amazon Web Services (AWS)

Santa Clara, CA

APPLIED SCIENTIST INTERN

Jun 2026 – Aug 2026

- Developed **ActObs** by extending agent SFT loss masks to supervise environment observations alongside actions without additional data or model parameters, and evaluated its impact on subsequent RL training for terminal agents.

University of Maryland, College Park

College Park, MD

GRADUATE ASSISTANT

Aug 2024 – Present

- Designed token-level teacher–student **logit fusion** for RL rollouts, combining importance-sampling correction with a decaying mixing schedule to enable a smooth transition from teacher guidance to student exploration.
- Developed frozen-projection adapters with task-specific sparse masks for model merging and continual learning (**LoRI**), reducing interference between tasks; LoRI outperformed LoRA with up to 95% fewer trainable parameters.

Tsinghua University

Beijing, China

RESEARCH ASSISTANT

Oct 2023 – May 2024

- Developed molecule tokenization for **UniMoT** using vector quantization and a Q-Former, representing molecules as discrete tokens for autoregressive molecule-to-text and text-to-molecule modeling.
- Co-developed atom-, motif-, and molecule-level graph tokenization for **HIGHT**, improving molecule-language alignment through hierarchical molecular representations; the method reduced hallucination by 40%.
- Contributed to automated search over subgraph selection and encoding strategies for drug-drug interaction prediction (**CSSE-DDI**).

Baidu Research

Beijing, China

RESEARCH INTERN

Mar 2023 – Oct 2023

- Developed **HL-GNN** to learn local and global link-prediction heuristics within a unified matrix formulation, using graph propagation and inter-layer connections for efficient computation.

Honors & Awards

2024	Dean's Fellowship	College Park, MD
2023	Outstanding Graduate of Shanghai Jiao Tong University	Shanghai, China
2021	Shanghai Government Scholarship	Shanghai, China
2021	Merit Student Award	Shanghai, China
2021	Academic Excellence Scholarship	Shanghai, China
2021	Meritorious Winner, International Mathematical Contest in Modeling (MCM)	USA

Service

Reviewer	NeurIPS 2025–2026; ICML 2024–2026; ICLR 2024–2027; KDD 2024; AAAI 2026–2027
Teaching Assistant	CMSC 250 Discrete Structures

Skills

Research	LLM post-training, agentic AI, RL, GRPO, SFT, PEFT/LoRA, self-improvement, continual learning
Frameworks	PyTorch, Hugging Face Transformers, vLLM, SGLang, verl, DeepSpeed, FSDP
Engineering	Python, C/C++, Ray, Docker, Git, Weights & Biases